# Diagnosis

## Clinical diagnosis: The importance of the neonatal period

The cutaneous lesions are major criteria for the diagnosis of incontinentia pigmenti (IP). Four stages are well characterized and summarized in Table 2, along with their main differential diagnoses.

- **Stage 1 (vesiculopustular stage):** This initial inflammatory stage is characterized by vesicles and pustules, typically appearing in the neonatal period. The lesions are often acral and follow Blaschko's lines. They may be profuse or limited in number. This stage is highly suggestive of IP when occurring in a female neonate with a linear, acral distribution and spontaneous resolution in early childhood. Eosinophilia is commonly observed during this stage; however, since eosinophilia can be physiological in early infancy, it is considered a minor criterion when associated with Stage 1 lesions.
- **Stage 2 (papular-verrucous stage):** This stage usually follows Stage 1 and presents as warty, hyperkeratotic papules or plaques, also distributed along Blaschko's lines. Histologically, it shows papillomatosis, acanthosis, hyperkeratosis, and numerous apoptotic keratinocytes in the epidermis, with dermal infiltration of eosinophils and lymphocytes.
- **Stage 3 (hyperpigmented stage):** Characterized by swirling, hyperpigmented macules that follow Blaschko's lines. These lesions typically appear after the resolution of earlier stages and may persist for years, gradually fading during adolescence.
- **Stage 4 (hypopigmented stage):** This late stage consists of linear, hypopigmented, and often hairless streaks or patches, predominantly on the

limbs. These lesions are permanent and may be subtle, sometimes requiring skin biopsy for confirmation.

Although IP predominantly affects females, it can rarely occur in males. The diagnostic criteria remain valid in male neonates, though the presentation may be more severe.

Spontaneous regression of Stages 1 and 2 is typical within the first months of life. However, late-onset outbreaks—linear and following Blaschko's lines—have been reported years after the neonatal period, sometimes triggered by viral infections. Therefore, IP should be considered in the differential diagnosis even in older children or adults with clinically suggestive skin lesions.

Histopathological examination is crucial, especially in atypical or late presentations. It is particularly important in cases of subungual verrucous lesions that are painful, to exclude benign or malignant nail tumors.

Cutaneous and scalp manifestations: - Alopecia may result from scarring due to inflammatory (Stage 1) or verrucous (Stage 2) lesions, typically on the vertex. These areas may be small and inconspicuous. - Hair, eyelashes, and eyebrows are often thin and sparse. Scalp hair may have a "woolly" texture, though without specific hair shaft abnormalities. - Nail abnormalities, such as striations or thickening, can occur but are non-specific and not pathognomonic.

## Skin histology

Histological findings support the clinical diagnosis and vary by stage:

- **Stage 1:** Eosinophilic spongiosis, intraepidermal vesicles containing eosinophils, dyskeratotic (apoptotic) keratinocytes, and a dermal infiltrate rich in eosinophils and lymphocytes.
- **Stage 2:** Papillomatosis, acanthosis, hyperkeratosis, and persistent apoptotic cells in the epidermis, with dermal eosinophilic infiltration.

The complete absence of minor criteria should prompt reconsideration of the diagnosis, as it may suggest an alternative condition.